Summary of Analytic Profiles, Variables, and Self-Similar Fits for the Subsonic (Ablation) and Shock Regimes

Compiled from: `piston_shock.pdf`, `sub (2).pdf`,
`bath_temperature_derivation (1).pdf`,
and `Eulerian_coordinate_derivation.tex`

Contents

1 Problem Setup **3**
1.1 Test Case Parameters (Gold) . . . 3

2 Self-Similar Coordinates and Exponents **3**
2.1 Subsonic (Ablation) Self-Similar Coordinate . . . 3
2.2 Self-Similar Profile Exponents . . . 3
2.3 Shock Self-Similar Coordinate . . . 4

3 Part 1: All Profiles in Both Regimes **4**
3.1 Subsonic (Ablation) Regime ($m < m_f(t)$) . . . 4
3.1.1 Specific Volume . . . 4
3.1.2 Density . . . 4
3.1.3 Velocity . . . 4
3.1.4 Pressure . . . 4
3.1.5 Temperature . . . 5
3.1.6 Specific Internal Energy . . . 5
3.1.7 Radiation Energy Flux . . . 5
3.2 Shock Regime ($m \geq m_f(t),\ \xi_{\text{shock}} \leq \xi_s$) . . . 5
3.2.1 Specific Volume . . . 5
3.2.2 Density . . . 5
3.2.3 Velocity . . . 5
3.2.4 Pressure . . . 6
3.2.5 Temperature (via EOS) . . . 6
3.2.6 Specific Internal Energy . . . 6
3.2.7 Radiation Energy Flux . . . 6
3.2.8 Hugoniot Jump Relations at the Shock Front ($\xi_s = \xi_s$) . . . 6
3.3 Unshocked Region ($\xi_{\text{shock}} > \xi_s$) . . . 6

4 Part 2: All Variables in Lagrangian and Eulerian Coordinates **6**
4.1 Subsonic (Ablation) Regime Variables . . . 6
4.1.1 Ablation Boundary ($m = 0$) . . . 6
4.1.2 Ablation (Heat) Front . . . 7
4.1.3 Distance from Boundary to Ablation Front . . . 7
4.1.4 Subsonic Region Position $x_{\text{sub}}(m, t)$ . . . 7

4.2 Shock Regime Variables . . . 7
4.2.1 Matching Conditions . . . 7
4.2.2 Shock Front . . . 7
4.2.3 Ablation Front Position (in the Patched Problem) . . . 8
4.2.4 Distance: Ablation Front to Shock Front . . . 8
4.2.5 Shock Region Position $x_{\text{shock}}(m, t)$ . . . 8
4.3 Energy Integrals . . . 8
4.3.1 Subsonic Regime . . . 8
4.3.2 Shock Regime ($\omega = 0$) . . . 9
4.4 Albedo and Surface Flux . . . 9

5 Part 2 (cont.): Numeric Values for the Two Test Cases 9
5.1 $\tau = 0$ (Constant Boundary Temperature) . . . 9
5.2 $\tau = 0.123$ (Constant Flux Temperature) . . . 10

6 Part 3: Self-Similar Profile Fits and Explicit Formulas 10
6.1 Subsonic Self-Similar Profile Fits . . . 10
6.1.1 $\tau = 0$ (Constant Boundary Temperature) . . . 10
6.1.2 $\tau = 0.123$ (Constant Flux Temperature) . . . 10
6.2 Substituting Fits into Dimensional Profiles . . . 11
6.2.1 Pressure $p(m, t)$ — Explicit Formula . . . 11
6.2.2 Temperature $T(m, t)$ — Explicit Formula . . . 11
6.2.3 Velocity $u(m, t)$ — Explicit Formula . . . 12
6.2.4 Specific Volume / Density — Explicit Formula . . . 12
6.2.5 Flux $s(m, t)$ — Explicit Formula . . . 12
6.3 Converting from $p(m, t)$ to $p(x, t)$ in the Subsonic Region . . . 12
6.4 Shock Regime: Profiles as Functions of x . . . 13

7 Shock Regime: Self-Similar Profile Structure 13

A Derivation of the Boundary Position 13

B Derivation of the Ablation Front Position Identity 14

C Derivation of the Closed-Form Position $x(m, t)$ 14

D Derivation of the Bath Temperature 14

E Shock V Integral Identity 14

F Energy Integral Identity 15

1 Problem Setup

One-temperature radiation-hydrodynamics in Lagrangian (m,t) coordinates with an ideal-gas EOS (γ, $r=\gamma-1$), power-law Rosseland opacity and specific internal energy:

$$p\,v = r\,e, \qquad \frac{1}{\kappa_R} = g\,T^{\alpha}\,v^{\lambda}, \qquad e = f\,T^{\beta}\,v^{\mu}. \tag{1}$$

Derived constants:

$$n = \frac{4+\alpha}{\beta}, \qquad k = 1-\mu, \qquad q = k\,n+\lambda-1. \tag{2}$$

Dimensional constants:

$$A = T_0\,(r\,f)^{1/\beta}, \qquad B = \frac{16\,\sigma_{\rm sb}\,g}{3\,\beta\,(r\,f)^{n}}, \qquad v_0 = \frac{1}{\rho_0}. \tag{3}$$

1.1 Test Case Parameters (Gold)

Parameter	$\tau=0$ (Fig 8)	$\tau=0.123$ (Fig 9)
α	1.5	1.5
β	1.6	1.6
λ	0.2	0.2
μ	0.14	0.14
γ	1.25	1.25
ω	0	0
ρ_0	19.32 g/cm^3	19.32 g/cm^3
T_0	1 HeV	1 HeV

2 Self-Similar Coordinates and Exponents

2.1 Subsonic (Ablation) Self-Similar Coordinate

$$\xi = m\,A^{a}\,B^{b}\,t^{c}, \tag{4}$$

with exponents:

$$a = \frac{2\beta-\beta\lambda-8-2\alpha+\mu(4+\alpha)}{4+2\lambda-4\mu}, \tag{5}$$

$$b = \frac{\mu-2}{4+2\lambda-4\mu}, \tag{6}$$

$$c = \frac{3\mu-2\lambda-2+\tau\big(2(\beta-\alpha-4)-\beta\lambda+\mu(4+\alpha)\big)}{4+2\lambda-4\mu}. \tag{7}$$

2.2 Self-Similar Profile Exponents

The physical fields factor as $h(m,t) = H(\xi)\,A^{a_i}\,B^{b_i}\,t^{c_i}$:

$$v(m,t) = V(\xi)\,A^{a_1}\,B^{b_1}\,t^{c_1}, \tag{8}$$

$$u(m,t) = U(\xi)\,A^{a_2}\,B^{b_2}\,t^{c_2}, \tag{9}$$

$$p(m,t) = P(\xi)\,A^{a_3}\,B^{b_3}\,t^{c_3}, \tag{10}$$

where the exponent relations are:

$$(a_2, b_2, c_2) = (a_1 - a,\ b_1 - b,\ c_1 - c - 1), \tag{11}$$

$$(a_3, b_3, c_3) = (a_1 - 2a,\ b_1 - 2b,\ c_1 - 2c - 2). \tag{12}$$

Additional derived profiles:

$$T(m,t) = \left(\frac{p\,v^{1-\mu}}{r\,f}\right)^{1/\beta}, \tag{13}$$

$$e(m,t) = \frac{p\,v}{r} = \frac{P(\xi)\,V(\xi)}{r}\,A^{a_1+a_3}\,B^{b_1+b_3}\,t^{c_1+c_3}, \tag{14}$$

$$\rho(m,t) = \frac{1}{v(m,t)} = \frac{1}{V(\xi)}\,A^{-a_1}\,B^{-b_1}\,t^{-c_1}. \tag{15}$$

2.3 Shock Self-Similar Coordinate

For the piston-driven shock with $\omega = 0$ (uniform initial density):

$$\xi_{\text{shock}} = m\sqrt{\frac{v_0}{p_{0,s}\,t^{\tau_s+2}}}, \tag{16}$$

where $p_{0,s}$ and τ_s are inherited from the subsonic solution (see §**??**).

3 Part 1: All Profiles in Both Regimes

3.1 Subsonic (Ablation) Regime ($m < m_f(t)$)

All profiles are expressed in Lagrangian coordinates unless otherwise noted. We define $y = \xi/\xi_f$ as the normalized self-similar coordinate.

3.1.1 Specific Volume

$$\boxed{v_{\text{sub}}(m,t) = V(\xi)\,A^{a_1}\,B^{b_1}\,t^{c_1}} \tag{17}$$

3.1.2 Density

$$\boxed{\rho_{\text{sub}}(m,t) = \frac{1}{V(\xi)}\,A^{-a_1}\,B^{-b_1}\,t^{-c_1}} \tag{18}$$

3.1.3 Velocity

$$\boxed{u_{\text{sub}}(m,t) = U(\xi)\,A^{a_2}\,B^{b_2}\,t^{c_2}} \tag{19}$$

3.1.4 Pressure

$$\boxed{p_{\text{sub}}(m,t) = P(\xi)\,A^{a_3}\,B^{b_3}\,t^{c_3}} \tag{20}$$

3.1.5 Temperature

$$\boxed{T_{\text{sub}}(m,t) = T(\xi)\,T_0\,t^{\tau} = \left(\frac{P(\xi)\,V(\xi)^{1-\mu}}{r\,f}\right)^{1/\beta} \cdot A^{(a_3+a_1(1-\mu))/\beta}\,B^{(b_3+b_1(1-\mu))/\beta}\,t^{(c_3+c_1(1-\mu))/\beta}} \tag{21}$$

In practice, from the ODE solver, $T(\xi) = (P(\xi)\,V(\xi)^{1-\mu})^{1/\beta}$ and $T(0) = 1$ by construction.

3.1.6 Specific Internal Energy

$$\boxed{e_{\text{sub}}(m,t) = \frac{P(\xi)\,V(\xi)}{r}\,A^{a_1+a_3}\,B^{b_1+b_3}\,t^{c_1+c_3}} \tag{22}$$

3.1.7 Radiation Energy Flux

$$\boxed{s_{\text{sub}}(m,t) = S(\xi)\,A^{a_S}\,B^{b_S}\,t^{c_S}} \tag{23}$$

where:

$$S(\xi) = -P^{n-1}(\xi)\,V^{q}(\xi)\left[V(\xi)\,P'(\xi) + k\,P(\xi)\,V'(\xi)\right] \tag{24}$$

and the flux power exponents are:

$$a_S = a + (q+1)\,a_1 + n\,a_3, \tag{25}$$

$$b_S = 1 + b + (q+1)\,b_1 + n\,b_3, \tag{26}$$

$$c_S = c + (q+1)\,c_1 + n\,c_3. \tag{27}$$

3.2 Shock Regime ($m \geq m_f(t)$, $\xi_{\text{shock}} \leq \xi_s$)

The shock regime uses a separate self-similar coordinate $\xi_s = m\sqrt{v_0/(p_{0,s}\,t^{\tau_s+2})}$ and self-similar profiles $V_s(\xi_s)$, $U_s(\xi_s)$, $P_s(\xi_s)$.

3.2.1 Specific Volume

$$\boxed{v_{\text{shock}}(m,t) = V_s(\xi_s)\,\left(v_0^2\,p_{0,s}^{\omega}\,t^{\omega(\tau_s+2)}\right)^{1/(2-\omega)}} \tag{28}$$

For $\omega = 0$:

$$v_{\text{shock}}(m,t) = V_s(\xi_s)\,v_0 \tag{29}$$

(i.e. V_s directly gives the ratio $v/v_0 = \rho_0/\rho$).

3.2.2 Density

$$\boxed{\rho_{\text{shock}}(m,t) = \frac{1}{V_s(\xi_s)\,v_0} \qquad (\omega = 0)} \tag{30}$$

3.2.3 Velocity

$$\boxed{u_{\text{shock}}(m,t) = U_s(\xi_s)\,\left(v_0\,p_{0,s}\,t^{\omega+\tau_s}\right)^{1/(2-\omega)}} \tag{31}$$

For $\omega = 0$:

$$u_{\text{shock}}(m,t) = U_s(\xi_s)\,\sqrt{v_0\,p_{0,s}}\,t^{\tau_s/2} \tag{32}$$

3.2.4 Pressure

$$\boxed{p_{\text{shock}}(m,t) = P_s(\xi_s)\, p_{0,s}\, t^{\tau_s}} \tag{33}$$

3.2.5 Temperature (via EOS)

$$\boxed{T_{\text{shock}}(m,t) = \left(\frac{p_{\text{shock}}\, \rho_{\text{shock}}^{\mu-1}}{r\, f}\right)^{1/\beta}} \tag{34}$$

3.2.6 Specific Internal Energy

$$\boxed{e_{\text{shock}}(m,t) = \frac{p_{\text{shock}}\, v_{\text{shock}}}{r} = \frac{P_s(\xi_s)\, V_s(\xi_s)\, p_{0,s}\, v_0}{r}\, t^{\tau_s}} \qquad (\omega = 0) \tag{35}$$

3.2.7 Radiation Energy Flux

$$\boxed{s_{\text{shock}}(m,t) = 0} \tag{36}$$

(Adiabatic region — no radiation transport.)

3.2.8 Hugoniot Jump Relations at the Shock Front ($\xi_s = \xi_s$)

For $\omega = 0$:

$$V_s(\xi_s) = \frac{r}{r+2}, \qquad U_s(\xi_s) = \frac{2(\tau_s+2)}{r(2-\omega)}\, \xi_s\, V_s(\xi_s), \qquad P_s(\xi_s) = \frac{r}{V_s(\xi_s)} \cdot \frac{U_s^2(\xi_s)}{2}. \tag{37}$$

At the piston ($\xi_s = 0$): $P_s(0) = 1$ by the shooting normalization.

3.3 Unshocked Region ($\xi_{\text{shock}} > \xi_s$)

$$v = v_0, \quad u = 0, \quad p = 0, \quad T = 0, \quad e = 0, \quad s = 0. \tag{38}$$

4 Part 2: All Variables in Lagrangian and Eulerian Coordinates

4.1 Subsonic (Ablation) Regime Variables

4.1.1 Ablation Boundary ($m = 0$)

Lagrangian: $m_b(t) = 0$ identically.

Eulerian position:

$$\boxed{x_b(t) = \frac{A^{a_2}\, B^{b_2}}{c_2+1}\, U(0)\, t^{c_2+1}} \tag{39}$$

Since $U(0) < 0$, the boundary moves in the $-x$ direction. This is a power law $x_b(t) = C_b\, t^{d_b}$ with:

$$C_b = \frac{A^{a_2}\, B^{b_2}}{c_2+1}\, U(0), \qquad d_b = c_2 + 1. \tag{40}$$

4.1.2 Ablation (Heat) Front

Lagrangian — ablated mass:

$$\boxed{m_f(t) = \xi_f\, A^{-a}\, B^{-b}\, t^{-c}} \tag{41}$$

This is a power law $m_f(t) = C_m\, t^{d_m}$ with $d_m = -c > 0$.

Eulerian position (standalone subsonic): $x_f(t) \equiv 0$ (by construction, the heat front is fixed at the origin in the pure subsonic problem).

Ablation front velocity: $\dot{x}_f(t) = 0$.

4.1.3 Distance from Boundary to Ablation Front

$$\boxed{\Delta x_f(t) = x_f(t) - x_b(t) = -x_b(t) = -\frac{A^{a_2}\, B^{b_2}}{c_2 + 1}\, U(0)\, t^{c_2+1} > 0} \tag{42}$$

Equivalently, from the integral formula:

$$\Delta x_f(t) = A^{a_2}\, B^{b_2}\, t^{c_2+1} \int_0^{\xi_f} V(\xi')\, d\xi' = -\frac{A^{a_2}\, B^{b_2}}{c_2 + 1}\, U(0)\, t^{c_2+1}, \tag{43}$$

using the integral identity $\int_0^{\xi_f} V\, d\xi = -U(0)/(c_2 + 1)$.

4.1.4 Subsonic Region Position $x_{\mathrm{sub}}(m, t)$

For $m < m_f(t)$:

$$\boxed{x_{\mathrm{sub}}(m, t) = x_{\mathrm{ablation}}(t) + \frac{A^{a_2}\, B^{b_2}\, t^{c_2+1}}{c_2 + 1} \Big[U\big(\xi(m)\big) - c\, \xi(m)\, V\big(\xi(m)\big) \Big]} \tag{44}$$

where $\xi(m) = m\, A^a\, B^b\, t^c$. In the standalone subsonic problem, $x_{\mathrm{ablation}}(t) = 0$; in the patched problem, it is the piston position of the shock solution (see §4.2).

4.2 Shock Regime Variables

4.2.1 Matching Conditions

The pressure at the ablation front provides the piston boundary condition for the shock:

$$\boxed{p_{0,s} = P_f\, A^{a_3}\, B^{b_3}, \qquad \tau_s = c_3.} \tag{45}$$

4.2.2 Shock Front

Lagrangian — shocked mass:

$$\boxed{m_s(t) = m_f(t) + \frac{\xi_s}{\sqrt{v_0/(p_{0,s}\, t^{\tau_s+2})}}} \qquad \text{(total mass up to the shock)} \tag{46}$$

The shocked-only mass (from piston to shock) for $\omega = 0$:

$$m_{\mathrm{shocked}}(t) = \xi_s \sqrt{\frac{p_{0,s}\, t^{\tau_s+2}}{v_0}}. \tag{47}$$

Eulerian — shock front position ($\omega = 0$):

$$\boxed{x_{\text{shock front}}(t) = \sqrt{v_0\, p_{0,s}}\; \xi_s\; t^{(\tau_s+2)/2}} \tag{48}$$

This is a power law $x_s(t) = C_s\, t^{d_s}$ with $d_s = (\tau_s + 2)/2$.

4.2.3 Ablation Front Position (in the Patched Problem)

Evaluated from the shock solution at $m = m_f(t)$:

$$\boxed{x_{\text{ablation}}(t) = \sqrt{v_0\, p_{0,s}}\; t^{(\tau_s+2)/2} \left[\hat{\xi}(t)\, V_s\big(\hat{\xi}(t)\big) + \frac{2}{\tau_s + 2}\, U_s\big(\hat{\xi}(t)\big)\right]} \tag{49}$$

where the time-dependent coordinate of the ablated mass in the shock frame is:

$$\hat{\xi}(t) = C_{\hat{\xi}}\, t^{d_{\hat{\xi}}}, \qquad C_{\hat{\xi}} = \xi_f\, A^{-a}\, B^{-b} \sqrt{\frac{v_0}{p_{0,s}}}, \qquad d_{\hat{\xi}} = -c - \frac{\tau_s + 2}{2}. \tag{50}$$

Not a simple power law

Since $d_{\hat{\xi}} \neq 0$ in general, the combined sub+shock problem is *not* fully self-similar. The ablation front position $x_{\text{ablation}}(t)$ involves the shock profiles evaluated at a time-varying argument and is therefore **not** a simple $C\, t^d$ power law.

4.2.4 Distance: Ablation Front to Shock Front

$$\boxed{\Delta x_{\text{shock}}(t) = x_{\text{shock front}}(t) - x_{\text{ablation}}(t) = \sqrt{v_0\, p_{0,s}}\; t^{(\tau_s+2)/2} \int_{\hat{\xi}(t)}^{\xi_s} V_s(\xi')\, d\xi'} \tag{51}$$

where the integral is computed via the exact analytic formula (for $\omega = 0$):

$$\int_{\xi_a}^{\xi_b} V_s\, d\xi = (1-\omega)\big(\xi_b V_s(\xi_b) - \xi_a V_s(\xi_a)\big) + \frac{1-\omega}{(\tau_s+2)/(2-\omega)}\big(U_s(\xi_b) - U_s(\xi_a)\big). \tag{52}$$

4.2.5 Shock Region Position $x_{\text{shock}}(m,t)$

For $m \geq m_f(t)$ and $\xi_s(m,t) \leq \xi_s$ ($\omega = 0$):

$$\boxed{x_{\text{shock}}(m,t) = \sqrt{v_0\, p_{0,s}}\; t^{(\tau_s+2)/2} \left[\xi_s(m,t)\, V_s\big(\xi_s(m,t)\big) + \frac{2}{\tau_s + 2}\, U_s\big(\xi_s(m,t)\big)\right]} \tag{53}$$

For m beyond the shock: $x(m,t) = v_0\, m$ (unperturbed).

4.3 Energy Integrals

4.3.1 Subsonic Regime

Total internal energy in ablated region:

$$\boxed{E_{\text{in,sub}}(t) = A^{2a_2 - a}\, B^{2b_2 - b}\, t^{2c_2 - c} \int_0^{\xi_f} \frac{P\, V}{r}\, d\xi} \tag{54}$$

Total kinetic energy in ablated region:

$$\boxed{E_{k,\mathrm{sub}}(t) = A^{2a_2-a}\, B^{2b_2-b}\, t^{2c_2-c} \int_0^{\xi_f} \frac{U^2}{2}\, d\xi} \tag{55}$$

Both share the time power t^{2c_2-c}, so E_k/E_{in} is constant.

4.3.2 Shock Regime ($\omega = 0$)

Total energy in shocked region:

$$\boxed{E_{\mathrm{total,shock}}(t) = \sqrt{v_0\, p_{0,s}^3}\; t^{\tau_s+1} \cdot \frac{U_s(0)}{\tau_s+1}} \tag{56}$$

(using the exact identity from piston shock theory).

4.4 Albedo and Surface Flux

Flux at $m = 0$ (surface/boundary) in the heat region:

$$\boxed{F_s(t) = s(0,t) = S_0\, A^{a_S}\, B^{b_S}\, t^{c_S}} \tag{57}$$

where $S_0 = S(\xi = 0)$ from the subsonic self-similar solution.

Albedo (ratio of outgoing flux to incoming blackbody flux):

$$\boxed{\alpha_{\mathrm{albedo}}(t) = \frac{2\, F_s(t)}{a_R\, c_{\mathrm{light}}\, T_s^4(t)} = \mathcal{B}_{\mathrm{bath}}\, t^{\eta_{\mathrm{bath}}}} \tag{58}$$

with:

$$\mathcal{B}_{\mathrm{bath}} = \frac{S_0\, A^{a_S}\, B^{b_S}}{2\, \sigma_{\mathrm{sb}}\, T_0^4}, \tag{59}$$

$$\eta_{\mathrm{bath}} = c_S - 4\,\tau. \tag{60}$$

Bath temperature (Marshak boundary condition):

$$\boxed{T_{\mathrm{bath}}(t) = T_0\, t^{\tau} \left[1 + \mathcal{B}_{\mathrm{bath}}\, t^{\eta_{\mathrm{bath}}}\right]^{1/4}} \tag{61}$$

5 Part 2 (cont.): Numeric Values for the Two Test Cases

5.1 $\tau = 0$ (Constant Boundary Temperature)

Self-similar constants: $\xi_f = 0.4277$, $P_f = 0.3871$, $U(0) = -10.544$, $S_0 \approx 1.603$.

Shock parameters: $p_{0,s} = 2.540 \times 10^8$ Ba, $\tau_s = -0.4479$, $\xi_s = 1.671$.

Quantity	Coefficient C_i	Time power d_i
$x_b(t)$	-6.411×10^7 cm/s^d	1.0365
$m_f(t)$	4.454×10^1 g cm^{-2}/s^d	0.5156
$x_{\mathrm{shock\,front}}(t)$	6.060×10^3 cm/s^d	0.7760
$\hat{\xi}(t)$	6.389×10^{-1} s^{-d}	$d_{\hat{\xi}} = -0.2604$

Bath drive: $\mathcal{B}_{\mathrm{bath}} \approx 0.0577$ (for t in ns), $\eta_{\mathrm{bath}} \approx -0.615$ (constant bath at $\tau = 0$).

5.2 $\tau = 0.123$ (Constant Flux Temperature)

Self-similar constants: $\xi_f = 0.3502$, $P_f = 0.4372$, $U(0) = -7.760$, $S_0 \approx 1.931$.
Shock parameters: $p_{0,s} = 2.335 \times 10^{11}$ Ba, $\tau_s = -0.1245$, $\xi_s = 1.177$.

Quantity	Coefficient C_i	Time power d_i
$x_b(t)$	-2.697×10^8 cm/s^d	1.1245
$m_f(t)$	4.787×10^3 g cm^{-2}/s^d	0.7510
$x_{\text{shock front}}(t)$	1.293×10^5 cm/s^d	0.9377
$\hat{\xi}(t)$	7.142×10^{-2} s^{-d}	$d_{\hat{\xi}} = -0.1868$

6 Part 3: Self-Similar Profile Fits and Explicit Formulas

The subsonic self-similar profiles $\tilde{T}$, $\tilde{P}$, $\tilde{U}$, $\tilde{S}$ are obtained numerically from the ODE solver. The following power-law fits were produced by `plot_test_cases.py`, expressed in the normalized variable $y = \xi/\xi_f \in [0, 1]$.

6.1 Subsonic Self-Similar Profile Fits

6.1.1 $\tau = 0$ (Constant Boundary Temperature)

Profile	Fit Formula	R^2
$\tilde{T}(y)$	$1.011\,(1-y)^{0.242}$	0.9996
$\tilde{P}(y)$	$0.355\,y^{0.902}$	0.9987
$\tilde{S}(y)$	$1.603\,(1-y)^{0.545}$	0.9975
Velocity fits (selected):		
$\tilde{U}_{\text{fit}\,3}(y)$	$-1.978\,(1-y)\,y^{-0.293}$	0.9950
$\tilde{U}_{\text{fit}\,4}(y)$	$-3.337\,(1-y^{0.516})\,y^{-0.190}$	0.9970
$\tilde{U}_{\text{fit}\,5}(y)$	$-1.964\,(1-y)\,(y^{8.473}+y^{-0.295})$	0.9958
$\tilde{U}_{\text{fit}\,7}(y)$	$-2.152\,(1-y)\,y^{-0.271}\,(1-0.194\,y)$	0.9958

6.1.2 $\tau = 0.123$ (Constant Flux Temperature)

Profile	Fit Formula	R^2
$\tilde{T}(y)$	$1.011\,(1-y)^{0.243}$	0.9997
$\tilde{P}(y)$	$0.394\,y^{0.909}$	0.9976
$\tilde{S}(y)$	$1.931\,(1-y)^{0.560}$	0.9972
Velocity fits (selected):		
$\tilde{U}_{\text{fit}\,3}(y)$	$-1.832\,(1-y)\,y^{-0.254}$	0.9947
$\tilde{U}_{\text{fit}\,4}(y)$	$-2.067\,(1-y^{0.846})\,y^{-0.227}$	0.9949
$\tilde{U}_{\text{fit}\,5}(y)$	$-1.798\,(1-y)\,(y^{5.988}+y^{-0.259})$	0.9971
$\tilde{U}_{\text{fit}\,7}(y)$	$-1.828\,(1-y)\,y^{-0.254}\,(1+0.005\,y)$	0.9947

Best velocity fit

The best R^2 for velocity is consistently Fit 4 ($c(1-y^a)y^{-b}$, 3 parameters) for $\tau = 0$, and Fit 5 ($c(1-y)(y^a + y^{-b})$, 3 parameters) for $\tau = 0.123$. Both achieve $R^2 > 0.997$.

6.2 Substituting Fits into Dimensional Profiles

The entire point of the self-similar fits is to produce **explicit closed-form** expressions for the full dimensional profiles. Below we show how this works for each profile, and then specialize to the two test cases.

6.2.1 Pressure $p(m,t)$ — Explicit Formula

From $p(m,t) = P(\xi)\, A^{a_3}\, B^{b_3}\, t^{c_3}$ and the fit $\tilde{P}_{\text{fit}}(y) = c_P\, y^{d_P}$:

$$\boxed{p(m,t) = c_P \left(\frac{m}{m_f(t)}\right)^{d_P} \cdot A^{a_3}\, B^{b_3}\, t^{c_3}} \tag{62}$$

Since $y = \xi/\xi_f = m/m_f(t)$, this becomes:

$$p(m,t) = c_P\, \xi_f^{-d_P}\, A^{a_3 + a\, d_P}\, B^{b_3 + b\, d_P}\, m^{d_P}\, t^{c_3 + c\, d_P} \tag{63}$$

which is an explicit power law in both m and t.

For $\tau = 0$: $c_P = 0.355$, $d_P = 0.902$:

$$p(m,t) = 0.355\, (0.4277)^{-0.902}\, A^{a_3 + 0.902\, a}\, B^{b_3 + 0.902\, b}\, m^{0.902}\, t^{c_3 + 0.902\, c} \tag{64}$$

For $\tau = 0.123$: $c_P = 0.394$, $d_P = 0.909$:

$$p(m,t) = 0.394\, (0.3502)^{-0.909}\, A^{a_3 + 0.909\, a}\, B^{b_3 + 0.909\, b}\, m^{0.909}\, t^{c_3 + 0.909\, c} \tag{65}$$

6.2.2 Temperature $T(m,t)$ — Explicit Formula

From the fit $\tilde{T}_{\text{fit}}(y) = c_T\, (1-y)^{d_T}$ and $T(m,t) = T(\xi)\, T_0\, t^{\tau}$:

$$\boxed{T(m,t) = c_T\, T_0\, t^{\tau} \left(1 - \frac{m}{m_f(t)}\right)^{d_T}} \tag{66}$$

For $\tau = 0$:

$$T(m,t) = 1.011\, T_0 \left(1 - \frac{m}{m_f(t)}\right)^{0.242} \qquad \text{(time-independent profile shape!)} \tag{67}$$

For $\tau = 0.123$:

$$T(m,t) = 1.011\, T_0\, t^{0.123} \left(1 - \frac{m}{m_f(t)}\right)^{0.243} \tag{68}$$

6.2.3 Velocity $u(m,t)$ — Explicit Formula

Using the best fit (Fit 4: $\tilde{U}_{\text{fit}}(y) = c_U\,(1 - y^{a_U})\,y^{-b_U}$):

$$\boxed{u(m,t) = c_U\,A^{a_2}\,B^{b_2}\,t^{c_2}\,\left(1 - \left(\frac{m}{m_f(t)}\right)^{a_U}\right)\left(\frac{m}{m_f(t)}\right)^{-b_U}} \tag{69}$$

For $\tau = 0$: $(c_U, a_U, b_U) = (-3.337, 0.516, 0.190)$:

$$u(m,t) = -3.337\,A^{a_2}\,B^{b_2}\,t^{c_2}\,\left(1 - \left(\frac{m}{m_f(t)}\right)^{0.516}\right)\left(\frac{m}{m_f(t)}\right)^{-0.190} \tag{70}$$

For $\tau = 0.123$: $(c_U, a_U, b_U) = (-2.067, 0.846, 0.227)$:

$$u(m,t) = -2.067\,A^{a_2}\,B^{b_2}\,t^{c_2}\,\left(1 - \left(\frac{m}{m_f(t)}\right)^{0.846}\right)\left(\frac{m}{m_f(t)}\right)^{-0.227} \tag{71}$$

6.2.4 Specific Volume / Density — Explicit Formula

From the velocity fit and the relation $V = (U' - c\,\xi\,V')/(c_1)$ from the ODE, no independent simple fit for V is produced. However, we can note that the density profile is implicitly determined by the pressure and temperature fits via the EOS:

$$\rho(m,t) = \left(\frac{r\,f\,T(m,t)^{\beta}}{e(m,t)}\right)^{1/(1-\mu)}, \qquad e(m,t) = \frac{p(m,t)}{\rho(m,t)\,r}. \tag{72}$$

So combining the p and T fits:

$$\boxed{v(m,t) = \left(\frac{p(m,t)}{r\,f\,T(m,t)^{\beta}}\right)^{-1/(1-\mu)}} \tag{73}$$

6.2.5 Flux $s(m,t)$ — Explicit Formula

From the fit $\tilde{S}_{\text{fit}}(y) = c_S\,(1-y)^{d_S}$:

$$\boxed{s(m,t) = c_S\,A^{a_S}\,B^{b_S}\,t^{c_S}\,\left(1 - \frac{m}{m_f(t)}\right)^{d_S}} \tag{74}$$

For $\tau = 0$: $s(m,t) = 1.603\,A^{a_S}\,B^{b_S}\,t^{c_S}\,\left(1 - m/m_f(t)\right)^{0.545}$
For $\tau = 0.123$: $s(m,t) = 1.931\,A^{a_S}\,B^{b_S}\,t^{c_S}\,\left(1 - m/m_f(t)\right)^{0.560}$

6.3 Converting from $p(m,t)$ to $p(x,t)$ in the Subsonic Region

Given the closed-form Eulerian coordinate (from Eq. (44)):

$$x_{\text{sub}}(m,t) = x_{\text{ablation}}(t) + \frac{A^{a_2}\,B^{b_2}\,t^{c_2+1}}{c_2+1}\Big[U\big(\xi(m)\big) - c\,\xi(m)\,V\big(\xi(m)\big)\Big], \tag{75}$$

one can (numerically) invert $x \mapsto m(x,t)$ and then substitute into $p(m,t)$ to get $p(x,t)$. Since the $x(m)$ mapping involves the self-similar profiles U and V, the inversion is generally not analytic, but the fits make it efficiently computable.

6.4 Shock Regime: Profiles as Functions of x

For $\omega = 0$, the shock position of a mass element is:

$$x_{\text{shock}}(m,t) = \sqrt{v_0\, p_{0,s}}\; t^{(\tau_s+2)/2} \left[\xi_s(m,t)\, V_s(\xi_s) + \frac{2}{\tau_s+2}\, U_s(\xi_s)\right]. \tag{76}$$

So any profile $h(m,t)$ can be expressed as $h(x,t)$ by inverting this mapping.

7 Shock Regime: Self-Similar Profile Structure

The shock self-similar profiles $V_s(\xi_s)$, $U_s(\xi_s)$, $P_s(\xi_s)$ are governed by a different set of ODEs (the adiabatic piston-shock self-similar equations). They satisfy:

- At the shock $\xi_s = \xi_s$: Rankine–Hugoniot relations (see above).
- At the piston $\xi_s \to 0$: $P_s(0) = 1$ (normalization), $V_s \to V_0^s$ (finite), $U_s \to U_0^s$ (finite).

These profiles are smooth and monotonic between the piston and the shock. The V integral (exact) is:

$$\int_0^{\xi_0} V_s(\xi)\, d\xi = (1-\omega)\, \xi_0\, V_s(\xi_0) + \frac{2-\omega}{\tau_s+2}\big(U_s(\xi_0) - U_s(0)\big). \tag{77}$$

Note on shock fits

Currently, no power-law fits are produced for the shock self-similar profiles V_s, U_s, P_s (the existing fitting pipeline in `plot_test_cases.py` only fits the subsonic profiles). Fits for the shock profiles can be added analogously using the piston shock solver output. As placeholders:

$$\tilde{V}_s^{\text{fit}}(z) = \tilde{V}_{s,\text{fit}}(z), \quad z = \xi_s/\xi_s, \tag{78}$$
$$\tilde{U}_s^{\text{fit}}(z) = \tilde{U}_{s,\text{fit}}(z), \tag{79}$$
$$\tilde{P}_s^{\text{fit}}(z) = \tilde{P}_{s,\text{fit}}(z). \tag{80}$$

Once these fits are determined, the same substitution procedure from §6.2 applies to produce explicit $p_{\text{shock}}(m,t)$ etc.

A Derivation of the Boundary Position

The boundary velocity is $u_b(t) = U(0)\, A^{a_2}\, B^{b_2}\, t^{c_2}$. Integrating:

$$x_b(t) = \int_0^t u_b(t')\, dt' = U(0)\, A^{a_2}\, B^{b_2} \int_0^t t'^{c_2}\, dt' = \frac{U(0)\, A^{a_2}\, B^{b_2}}{c_2+1}\, t^{c_2+1}, \tag{81}$$

requiring $c_2 > -1$ for convergence.

B Derivation of the Ablation Front Position Identity

The Eulerian position of a Lagrangian element m is:

$$x(t,m) - x_b(t) = \int_0^m v(m',t)\,dm' = A^{a_2}\,B^{b_2}\,t^{c_2+1}\int_0^{\xi(m)} V(\xi')\,d\xi'. \tag{82}$$

At $m = m_f(t)$, the integral identity $\int_0^{\xi_f} V\,d\xi = -U(0)/(c_2+1)$ ensures $x_f(t) = 0$.

Proof: From the self-similar continuity equation $c\,\xi\,V' + c_1\,V - U' = 0$:

$$\int_0^{\xi_0} V\,d\xi = \frac{1}{c_1 - c}\left[U(\xi_0) - U(0) - c\,\xi_0\,V(\xi_0)\right]. \tag{83}$$

For $\xi_0 = \xi_f$ where $U(\xi_f) = V(\xi_f) = 0$: $\int_0^{\xi_f} V\,d\xi = -U(0)/(c_1 - c) = -U(0)/(c_2+1)$.

C Derivation of the Closed-Form Position $x(m,t)$

Starting from:

$$x(t,m) = A^{a_2}\,B^{b_2}\,t^{c_2+1}\left[\frac{U(0)}{c_2+1} + \int_0^{\xi(m)} V\,d\xi\right], \tag{84}$$

and using the general integral identity:

$$\int_0^{\xi_0} V\,d\xi = \frac{1}{c_2+1}\left[U(\xi_0) - U(0) - c\,\xi_0\,V(\xi_0)\right], \tag{85}$$

we get:

$$x(t,m) = \frac{A^{a_2}\,B^{b_2}\,t^{c_2+1}}{c_2+1}\left[U(\xi) - c\,\xi\,V(\xi)\right]. \tag{86}$$

D Derivation of the Bath Temperature

The Marshak boundary condition gives:

$$T^4_{\mathrm{bath}}(t) = T_s^4(t) + \frac{F_s(t)}{2\,\sigma_{\mathrm{sb}}}. \tag{87}$$

Using $F_s(t) = S_0\,A^{a_S}\,B^{b_S}\,t^{c_S}$:

$$T_{\mathrm{bath}}(t) = T_0\,t^{\tau}\left[1 + \frac{S_0\,A^{a_S}\,B^{b_S}}{2\,\sigma_{\mathrm{sb}}\,T_0^4}\,t^{c_S - 4\tau}\right]^{1/4}. \tag{88}$$

E Shock V Integral Identity

From the piston-shock self-similar continuity equation (for $\omega = 0$):

$$\int_0^{\xi_0} V_s\,d\xi = \xi_0\,V_s(\xi_0) + \frac{2}{\tau_s+2}\big(U_s(\xi_0) - U_s(0)\big). \tag{89}$$

This gives the piston position: $x_{\mathrm{piston}}(t) = \sqrt{v_0\,p_{0,s}}\,t^{(\tau_s+2)/2}\cdot U_s(0)\cdot 2/(\tau_s+2)$.

F Energy Integral Identity

The total energy integral satisfies (subsonic regime):

$$\int_0^{\xi_0} \left(\frac{PV}{r} + \frac{U^2}{2} \right) d\xi = \frac{1}{2c_2 - c} \left[-S(\xi) - P(\xi)\,U(\xi) - c\,\xi \left(\frac{PV}{r} + \frac{U^2}{2} \right) \right]_0^{\xi_0}. \tag{90}$$

At $\xi_0 = \xi_f$ (where $P = V = U = S = 0$):

$$E_{\text{total}} = \frac{1}{2c_2 - c} \left[S(0) + P(0)\,U(0) \right] = \frac{S_0}{2c_2 - c} \tag{91}$$

(since $P(0) = 0$ in the subsonic problem).